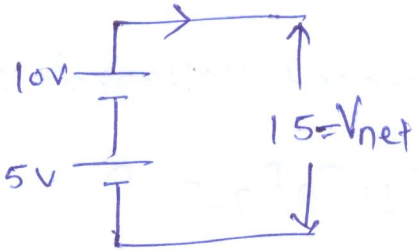
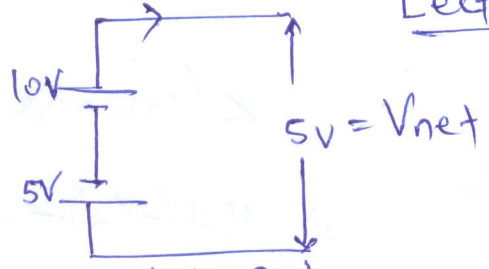
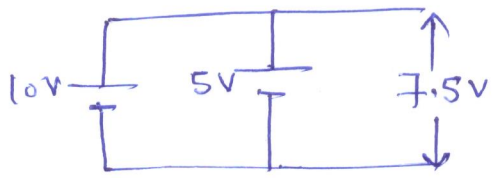
Lecture-02



(Fig-1)



(Fig-2)



→ considering unit resistance in each branches

Fig-3

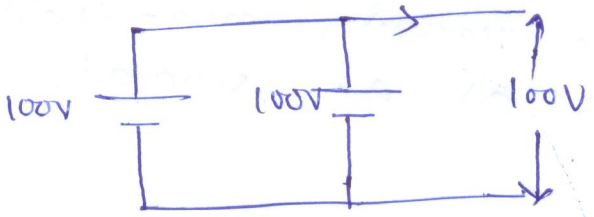
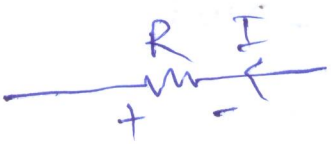


Fig-4

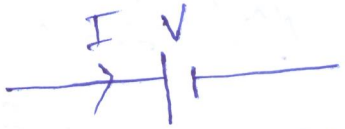
How to determine sign of voltage & voltage drop

① 

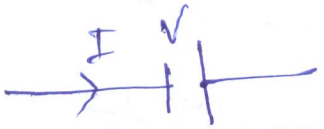
$V_R = -IR$

② 

$V_R = +IR$

③ 

Voltage = -V

④ 

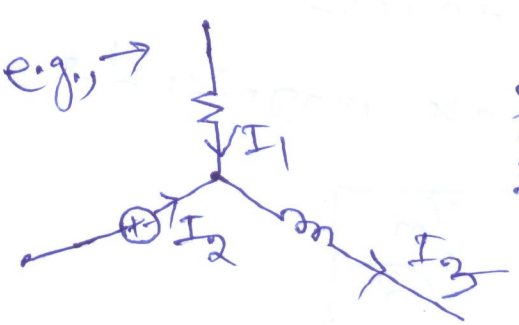
Voltage = +V

KCL →

At a node sum of all current equal to zero.

i.e $\sum_{i=1}^{N} I_i = 0$

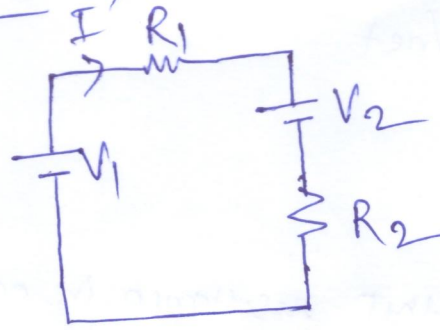
e.g., →



$\sum_{i=1}^{3} I_i = 0$

$\Rightarrow I_1 + I_2 = I_3$

KVL →



Σ (Voltage source + Voltage drop) = 0

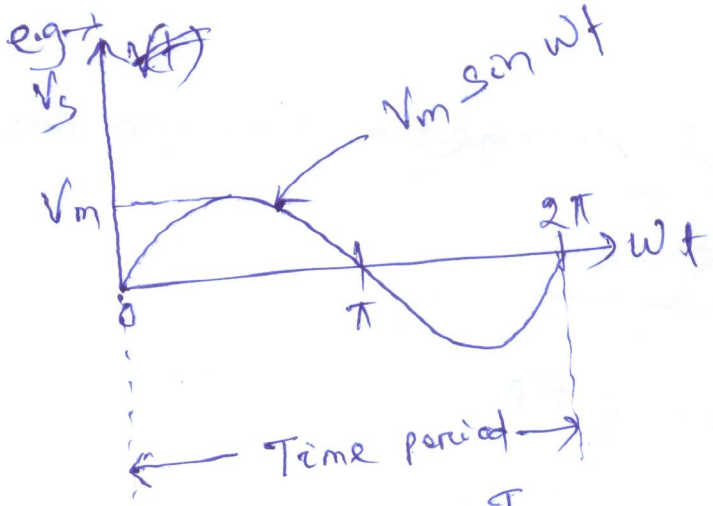
$\Rightarrow V_1 + (-V_2) - IR_1 - IR_2 = 0$

$\Rightarrow V_1 - V_2 - IR_1 - IR_2 = 0$

AC signal →

→ Denoted by (V)

→ The signal which have varying magnitude w.r.t. to time is known as AC signal.



→ Total time = T

→ frequency (f) = $\frac{1}{T}$ i.e $f = \frac{1}{T} = \frac{\text{No. of revolution}}{\text{Time}}$

→ $2\pi f = \omega$, ω → angular frequency (rad/sec)

$f = \frac{\omega}{2\pi}$

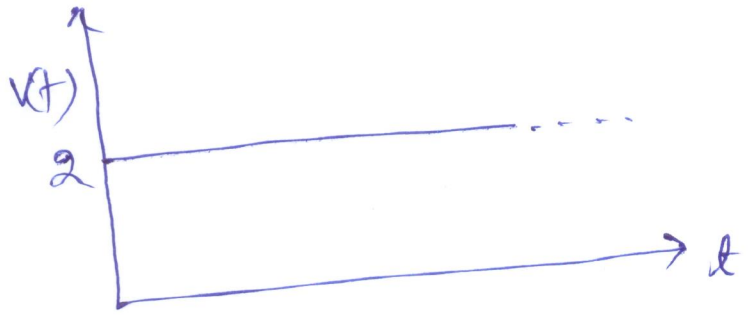
↳ in (HZ)

V_m → peak magnitude

$$V_{rms} = \frac{V_m}{\sqrt{2}}$$

dc signal :-

→ dc → direct current.

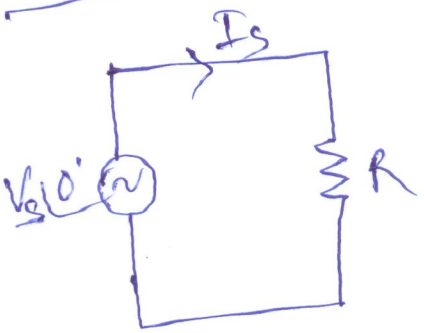


→ $f = 0$

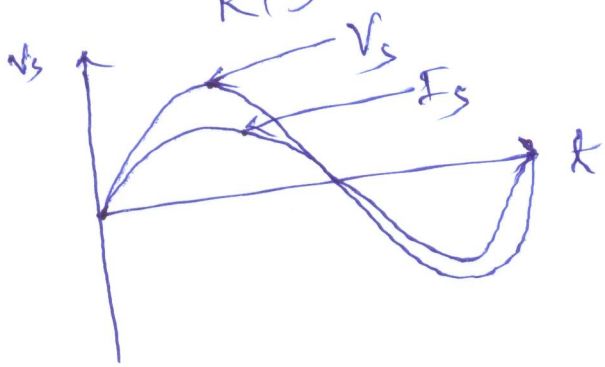
→ constant magnitude with respect to time.

→ symbol → ⊣⊢ or ⊣+⊢

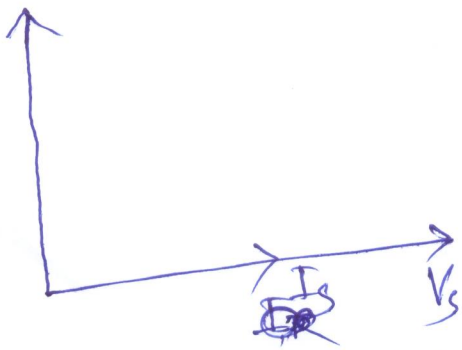
AC in pure Resistive element



$$I_s = \frac{V_s \angle 0^\circ}{R + j0} = \frac{V_s}{R} \angle 0 = I_m \angle 0^\circ$$



Phasor diagram :-



i.e., V_s & I_s coincide to eachother as there is no phase difference.